

Istanbul Atlas University

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TECHNOLOGY ACQUISITION AND TRANSFER STRATEGY FOR UBISOFT: AI-POWERED AUTOMATED GAME TESTING

PROJECT REPORT Project No: 1

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1. INTRODUCTION

Technology plays an important role in the video game industry, where companies continuously develop complex games with advanced graphics, interactive environments, and diverse gameplay mechanics. As games become more sophisticated, effective testing becomes increasingly important. Identifying and fixing software bugs helps maintain game quality, improve the player experience, and prevent development delays.

This report focuses on Ubisoft, a well-known video game developer and publisher, and examines the challenges of testing large and complex games. Testing thousands of missions, characters, and gameplay mechanics requires considerable time and resources. Manual testing alone may not identify every potential problem efficiently, especially when bugs occur under specific conditions.

To address this challenge, the report proposes an artificial intelligence (AI)-powered automated game-testing system. AI agents would play selected missions, identify potential bugs, and generate reports for developers. The recommended acquisition strategy combines internal R&D by Ubisoft La Forge with a strategic technology partnership to leverage specialized external capabilities.

The report also evaluates technology acquisition alternatives and examines corporate research and development (R&D), technology transfer, intellectual property, risk management, and implementation planning. Its objective is to recommend a strategy that could improve Ubisoft's testing efficiency, reduce development costs, and enhance game quality.

2. ORGANIZATION AND INDUSTRY BACKGROUND

Ubisoft is a French video game company founded in 1986. It develops and publishes games for computers and gaming consoles and operates internationally. Its well-known franchises include *Assassin's Creed*, *Far Cry*, *Just Dance*, and *Tom Clancy's Rainbow Six* (Ubisoft, n.d.).

The video game industry combines software engineering, creative design, and advanced technologies. Modern games feature detailed environments, interactive characters, complex

missions, and diverse gameplay mechanics. Developing these features requires collaboration among programmers, game designers, animators, and quality assurance teams.

Software testing is essential for identifying bugs and correcting technical problems before release. However, complex games must function correctly across numerous missions, characters, and gameplay scenarios. As games grow in size and complexity, ensuring their quality requires increasing time and resources.

Automation and AI offer opportunities to improve testing. Automated tools can repeat specific actions, while AI-based systems can explore selected scenarios and identify potential problems. For Ubisoft, these technologies could support quality assurance teams and improve testing efficiency.

3. IDENTIFICATION OF THE TECHNOLOGY NEED

3.1 The Technology Problem

Developing large video games requires extensive testing to ensure that missions, characters, and gameplay mechanics function correctly. Ubisoft's complex games may contain thousands of missions, numerous character interactions, and different gameplay scenarios. Testing these elements manually requires considerable time and effort.

Some bugs occur only under specific conditions, making them difficult to detect. Developers must also repeat tests after fixing defects or updating features, increasing costs and potentially delaying releases.

Therefore, Ubisoft needs a more efficient testing approach that maintains game quality while automating repetitive tasks and helping developers identify bugs earlier.

3.2 The Proposed Technology Solution

This report proposes an AI-powered automated game-testing system. AI agents would interact with game environments, play selected missions, and test different gameplay scenarios automatically.

The agents would identify unusual behavior, such as characters becoming stuck, missions failing to progress, or mechanics not functioning as expected. The system would generate reports describing potential bugs, helping developers investigate and reproduce them.

Under the proposed hybrid model, Ubisoft will leverage internal development through Ubisoft La Forge alongside a strategic technology partnership. External tools and specialized frameworks will be integrated to support engine compatibility and specialized testing tasks while internal teams focus on core mechanics and player experience.

3.3 Expected Benefits

The proposed system could provide four main benefits.

First, automated agents could repeatedly test selected missions and scenarios, reducing the time required for repetitive tasks. Second, earlier defect detection could help developers resolve problems before release. Third, automation could reduce the effort and costs associated with repeated testing, although initial development and integration would require investment. Finally, improved test coverage could help identify technical problems and enhance game quality.

Overall, these benefits could lead to more efficient testing and fewer undetected bugs, provided that the system performs reliably in practice.

4. TECHNOLOGY ACQUISITION ALTERNATIVES

Ubisoft could obtain the technology through internal development, purchasing, licensing, partnerships, or acquisition. Each alternative differs in cost, implementation speed, technical capabilities, and long-term control.

4.1 Developed Internally (Make/R&D)

Ubisoft could develop its own AI-powered testing system using internal engineers and R&D teams. This would allow the company to tailor the technology to its games and development processes while maintaining greater control over its intellectual property.

However, internal development requires skilled employees, time, and significant initial investment.

4.2 Purchase Technology (Buy)

Ubisoft could purchase an existing automated testing solution from an external provider. This could accelerate implementation without requiring the company to build every component from scratch.

However, the product might not fully meet Ubisoft's requirements, and customization or integration could increase costs.

4.3 Licence External Technology

Licensing would allow Ubisoft to use existing AI testing tools without developing every component internally. This could reduce development time and provide access to specialized capabilities.

However, ongoing fees and dependence on providers for updates or technical support could limit flexibility.

4.4 Establish a Partnership

Ubisoft could collaborate with specialized AI testing companies or research institutions to develop an AI-assisted game-testing system. External partners could provide complementary testing tools, automation capabilities, integration support, and technical training, while Ubisoft La Forge contributes internal AI expertise and game-development knowledge.

However, partners would need to coordinate responsibilities, costs, and intellectual property ownership. The suitability of external technologies for Ubisoft's game engines and testing requirements would also need to be validated through a joint pilot project.

4.5 Acquire a Technology Company

Ubisoft could acquire a company specializing in AI testing, gaining access to existing technology, experienced employees, and technical expertise.

However, acquisitions require substantial financial resources and may create organizational and technical integration challenges. Therefore, this option could be more expensive than the alternatives.

5. EVALUATION AND SELECTION OF THE BEST ALTERNATIVE

The alternatives can be evaluated using five criteria: cost, speed, technical capability, strategic value, and long-term control. The criteria are weighted based on their strategic importance to Ubisoft: Technical Capability (30%), Strategic Value (25%), Cost (15%), Speed (15%), and Long-Term Control (15%).

Scores range from 1 to 5, where 1 indicates very low performance and 5 indicates very high performance. The scores represent a comparative assessment of the alternatives included in the matrix.

5.1 Technology Acquisition Matrix

The scores in the matrix are based on our comparative evaluation of each technology acquisition alternative.

Evaluation Criteria	**Weight**	**Make/R&D**	**Buy**	**License**	**Partnership**	**Acquire**
Cost	15%	2	3	4	3	1
Speed	15%	2	5	5	4	2
Technical Capability	30%	4	2	2	4	3
Strategic Value	25%	5	2	2	5	4
Long-Term Control	15%	5	2	2	4	5
Weighted Total (out of 5)	**100%**	**3.80**	**2.60**	**2.75**	**4.10**	**3.10**

5.2 Selection of the Best Alternative

Based on the weighted evaluation matrix, the Strategic Technology Partnership option receives the highest score of 4.10 out of 5. This result reflects the critical importance of technical capability and strategic value in the proposed acquisition strategy.

A partnership-led approach allows Ubisoft to combine its internal R&D capabilities, particularly through Ubisoft La Forge, with the specialized expertise and tooling of an external technology partner. Ubisoft contributes internal AI research, game-development experience, and engine integrations, while the external partner delivers automated testing frameworks, integration support, and technical training.

Before wider deployment, the partners will conduct a joint pilot project using selected game scenarios to evaluate engine compatibility, testing effectiveness, and intellectual property safeguards. Therefore, the recommended strategy is to prioritize a Strategic Technology Partnership supported by internal R&D.

6. CORPORATE R&D REQUIREMENTS

6.1 Role of Corporate R&D

Corporate R&D helps companies strengthen their technological capabilities, develop innovative solutions, and maintain competitive advantages. For Ubisoft, R&D could improve both technological innovation and the efficiency of game testing.

Because manually testing thousands of missions and gameplay mechanics is time-consuming, the proposed R&D strategy aims to increase test coverage, accelerate testing, and identify defects earlier through AI and automation.

Ubisoft's R&D unit, La Forge, connects research with game production. It provides a relevant example of how the company could investigate and adapt emerging technologies to its development processes.

6.2 Proposed R&D Activities

6.2.1 Reinforcement Learning-Based Autonomous Game Testing

Reinforcement learning enables AI agents to learn by interacting with a game environment. Agents could attempt missions, interact with characters, and explore unexpected behaviors. Ubisoft's R&D team could train agents across different scenarios to expand test coverage and reduce repetitive manual work. However, human testers would remain necessary because agents cannot represent every possible player behavior.

6.2.2 Automated Regression and Model-Based Testing

Regression testing checks whether software updates introduce defects into previously functioning features. Model-based testing uses representations of expected behavior to generate test scenarios.

Ubisoft could integrate these methods into its development workflows to verify critical missions and features automatically after updates.

6.2.3 Machine Learning-Based Bug Prediction and Anomaly Detection

Machine learning models can analyze historical defect reports and testing results to identify components at higher risk of failure. Anomaly detection can highlight unusual gameplay behavior or performance issues.

These methods could help testing teams prioritize high-risk areas. However, model performance should be evaluated regularly, and human testers should validate the findings.

6.3 R&D Strategy and Performance Measurement

A hybrid strategy combining internal R&D with an external technology partnership is recommended for Ubisoft. La Forge could build on its AI and machine learning research to develop and validate AI-assisted testing capabilities, while a potential external partner could contribute complementary testing tools, automation, and integration support.

R&D performance should be measured through testing time, test coverage, defect detection rates, false-positive rates, and integration effort. The proposed system should first be evaluated

through a joint pilot project. Wider implementation should proceed only if the results demonstrate sufficient testing effectiveness, engine compatibility, and acceptable costs.

7. TECHNOLOGY TRANSFER STRATEGY

7.1 Purpose of Technology Transfer

Technology transfer involves introducing technological knowledge and tools into an organization for practical use. It includes software acquisition, documentation, training, and integration support.

For Ubisoft, the objective is to integrate AI-based testing tools into existing game-development and quality assurance processes while ensuring compatibility with its game engines.

7.2 Recommended Technology Transfer Mechanism

A hybrid technology transfer model combining Ubisoft's internal R&D capabilities with a strategic technology partnership is recommended. Under this model, La Forge would contribute AI research and game-testing expertise, while the external partner would provide complementary testing tools, automation, integration support, and technical training.

The main advantages of this model are:

- **Access to Expertise:** External partners contribute specialized knowledge and cuttingedge testing capabilities.
- **Knowledge Transfer:** Joint experimentation, training, and documentation strengthen Ubisoft's absorptive capacity (Cohen & Levinthal, 1990) and internal technical expertise.
- **Integration Support:** Collaboration facilitates engine-level compatibility checks with proprietary game engines (such as Anvil and Snowdrop).
- **Maintaining Control:** Clear contractual frameworks protect Ubisoft's critical technology, confidential data, and intellectual property.

7.3 Implementation Stages

Technology transfer is organized into four core implementation phases matching the project roadmap:

1. **Scope & Select (Months 1–3):** Define pilot scenarios, establish baseline QA metrics, and select a strategic technology partner.
2. **Co-Develop & Pilot (Months 4–6):** Co-develop testing frameworks, build initial prototypes, and test engine compatibility on selected game builds.
3. **Validate & Integrate (Months 7–9):** Conduct joint evaluations between La Forge and QA teams, validate defect accuracy, train staff, and transfer technical knowledge.
4. **Selective Rollout (Months 10–12):** Expand validated tools to approved QA workflows and monitor ongoing performance.

7.4 Intellectual Property and Risk Management

Ubisoft must protect confidential information, including source code, unreleased content, and character designs. Agreements with external providers should define software usage rights, intellectual property ownership, confidentiality, and technical support responsibilities.

Licensing costs and dependence on external providers should also be managed. Developing internal expertise and using standard interfaces could reduce these risks. AI-generated findings should be validated by human testers.

The proposed model would enable Ubisoft to benefit from external expertise while building internal knowledge and integrating new testing technologies effectively.

8. RISK AND INTELLECTUAL PROPERTY ANALYSIS

8.1 Risk Analysis

AI agents could improve testing speed, defect detection, and cost efficiency. However, implementation involves technical, financial, operational, and security risks that Ubisoft must manage.

8.1.1 Technical Risks

AI agents may miss bugs or generate false positives. For example, an agent might interpret an intentional game mechanic as an error or fail to detect a defect that occurs only under specific conditions. Complex missions and frequent updates may also affect performance.

Ubisoft could reduce these risks by testing the system against known defects, validating reports through human testers, and combining AI-based testing with traditional methods.

8.1.2 Financial Risks

Internal development requires investment in engineers, AI specialists, computing infrastructure, licenses, and training. The system may also take longer than expected to deliver measurable benefits.

Ubisoft should begin with a small pilot and compare implementation and operating costs against improvements in testing efficiency before expanding the system.

8.1.3 Operational and Organizational Risks

Integration may disrupt established testing workflows, while employees may need time to learn the new technology.

Training, clear procedures, and collaboration between researchers, developers, and quality assurance teams could reduce these challenges. Human testers should continue evaluating complex defects and overall gameplay experience.

8.1.4 Data Security and Confidentiality Risks

External AI platforms and testing services may expose sensitive information, including source code, unreleased content, and proprietary development tools.

Ubisoft should assess providers' security practices, restrict access to confidential data, and establish contractual confidentiality requirements. Game assets should only be shared externally when necessary and authorized.

8.2 Intellectual Property Analysis

Intellectual property (IP) protection is important because the proposed system may contain valuable software, AI models, testing methods, and proprietary game data.

Under the recommended strategy, Ubisoft relies on a strategic technology partnership supported by internal R&D (La Forge). Ownership of internally developed software, algorithms, and core game telemetry remains strictly with Ubisoft.

Partnership agreements must explicitly specify permitted usage rights, restrictions on codeveloped assets, confidentiality obligations, and technical support terms. For joint R&D initiatives, written contracts will clarify ownership of IP outputs, research data access, and commercialization rights to ensure Ubisoft maintains full long-term strategic control.

9. IMPLEMENTATION ROADMAP

9.1 Implementation Approach

Ubisoft should introduce the AI-powered testing system in four structured phases over a 12month period. This phased approach limits financial investment until technical viability and engine compatibility are validated.

Phase 1 — Scope & Select (Months 1–3):

Define pilot scenarios, establish baseline QA metrics, and select a potential technology partner.

Phase 2 — Co-Develop & Pilot (Months 4–6):

Assess engine compatibility, test accuracy, security, and CI/CD integration.

Phase 3 — Validate & Integrate (Months 7–9):

La Forge, QA teams, and the technology partner test AI agents on a selected game build.

Phase 4 — Selective Rollout (Months 10–12):

Expand to approved QA workflows, train teams, and monitor performance.

9.2 Implementation Timeline

The following timeline is a proposed planning estimate, not an official Ubisoft project schedule.

Phase	Estimated duration	Main activity
1. Scope & Select	Months 1–3	Define pilot scope, establish baseline QA metrics, and select technology partner
2. Co-Develop & Pilot	Months 4–6	Co-develop AI testing agents and run pilot tests on selected game builds
3. Validate & Integrate	Months 7–9	Evaluate AI accuracy, integrate with CI/CD pipelines, and train QA teams.
4. Selective Rollout	Months 10–12	Expand validated system to production workflows and measure ROI.

Activities may overlap depending on resource availability and development requirements. The timeline should be revised after the feasibility assessment.

9.3 Key Performance Indicators

Ubisoft should use measurable performance indicators to determine whether the system achieves its objectives.

Performance Indicator	Measurement Purpose
Testing Time	Measures changes in the time required to complete selected tests.
Defect Detection Rate	Measures the system's ability to identify valid software defects
Test Coverage	Tracks the percentage of priority gameplay scenarios tested automatically.
QA-Confirmed Finding Rate	Measures the percentage of AI-reported issues confirmed as valid defects by human QA testers.
Regression Testing Time	Measures the execution speed of repetitive tests following game updates.
Testing Cost	Compare the cost of the new system with existing testing processes
Employee Adoption	Tracks training completion and regular system usage

Baseline values should be collected before the pilot begins. Improvement targets should then be established using the results rather than assumed in advance.

The system should be expanded only if it demonstrates reliable defect detection, meaningful efficiency improvements, acceptable operating costs, and compatibility with human quality assurance processes.

10. CONCLUSION

The increasing complexity of modern video games creates significant challenges for software testing and quality assurance. For Ubisoft, manually testing large numbers of missions, characters, and gameplay interactions requires substantial time, effort, and financial resources. AI-powered automated testing offers a scalable solution to address this challenge by enabling intelligent agents to explore complex game environments, repeat regression scenarios, and report potential defects early in the development lifecycle.

This report recommends a Strategic Technology Partnership supported by Ubisoft's internal R&D capabilities through Ubisoft La Forge. Under this hybrid approach, La Forge will drive core machine learning research and game engine integrations (such as Anvil and Snowdrop), while an external technology partner provides complementary testing frameworks, automation tools, and technical training.

To mitigate technical, financial, operational, and security risks, implementation will follow a 12-month, 4-phase rollout plan (Scope & Select, Co-Develop & Pilot, Validate & Integrate, and Selective Rollout). A strict Go/No-Go decision gate will ensure that full-scale adoption proceeds only after pilot testing confirms engine compatibility, high defect detection accuracy, acceptable false-positive rates, and strong intellectual property safeguards.

Ultimately, combining internal R&D innovation with an external strategic technology partnership allows Ubisoft to build robust, long-term testing capabilities. This strategy could increase test coverage, reduce repetitive QA workload, accelerate feedback loops, and lower development costs—ensuring high-quality and reliable gaming experiences for players worldwide.

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